CS23E01 / EMBEDDED SYSTEMS AND INTERNET OF THINGS

- Session: July-Nov 2026 (B.E. CSE – VII sem)

Activities for Lab-3:

Topics: I/O operations, rotate and serialization, Logic and compare Operations, Timers, Delays and Counters

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- Connect a switch to the LEDs and on the flip of a switch alternately display the patterns AAH and 55H.
- Read the status of switches and display the same on LEDs.
- Read an 8-bit input and display its complement on the switch.
- Rotate an 8-bit data left continuously and display the result on LEDs.
- Generate a circular LED pattern using rotate instructions.
- Serialize the data byte A5H through a single output pin.
- Complement an 8-bit number and display the result.
- Compare two 8-bit numbers and display the result using three LEDs. Turn ON the first LED if the first number is greater than the second, turn ON the second LED if the second number is greater than the first, and turn ON the third LED if both numbers are equal. Ensure that only one LED is ON for each comparison.
- Determine whether an input number is even or odd using logical operations.
- Check whether a specified bit in a byte is set or cleared and indicate the result using an LED.
- Blink an LED at 1-second intervals using Timer 0.
- Design a traffic light controller using LEDs and timer delays.

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- Install SDCC, MCU8051 IDE, and experiment with simple C to just begin your next session of programs in embedded C.